

SMART INDIA HACKATHON 2026

PROJECT REPORT

AI-Powered Smart Onion Storage Monitoring & Environmental Control System

“Onichan”

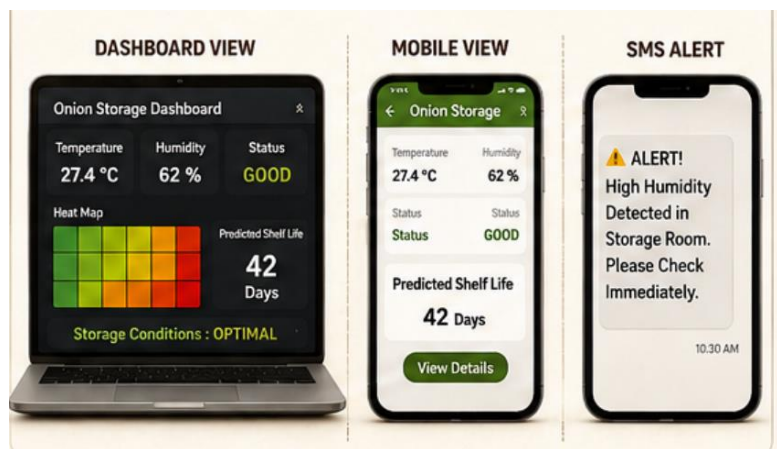


Fig. 0.1 — Onichan smart onion storage dashboard, mobile view and SMS alert concept

Field	Details
Problem Statement Title	Improved Onion Storage Technology for Enhancing Shelf Life of Onion
Theme	Agriculture, FoodTech & Rural Development
PS Category	Hardware
Team Name (registered on portal)	Onichan
Team ID	119323
Institution	K. C. College of Engineering & Management Studies & Research

Certificate

K. C. College of Engineering & Management Studies & Research

This is to certify that the project report entitled “AI-Powered Smart Onion Storage Monitoring & Environmental Control System (Onichan)” is a bona fide record of work carried out by the team members whose names appear on the following Project Approval Sheet, in partial fulfilment of the requirements of the Smart India Hackathon 2026, under the theme Agriculture, FoodTech & Rural Development.

Date: 19/09/2026 Place: Thane

Project Approval Sheet

The project report entitled “AI-Powered Smart Onion Storage Monitoring & Environmental Control System (Onichan)”, submitted for Smart India Hackathon 2026, is hereby approved as a credible piece of technical work.

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Declaration

We hereby declare that the project report entitled “AI-Powered Smart Onion Storage Monitoring & Environmental Control System (Onichan)”, submitted for Smart India Hackathon 2026, is our own original work carried out under the guidance of our mentor(s). The content presented is based on our team's design, the cited literature and publicly available research, and has not been submitted elsewhere for the award of any degree or prize.

Team Name: Onichan

Date: 19/09/2026 Place: Thane

Acknowledgement

We would like to express our sincere gratitude to our mentors, faculty coordinators, and the Smart India Hackathon 2026 organizing committee for their continuous guidance and encouragement throughout the development of this project.

We also thank the researchers and institutions — including ICAR-DOGR, NHRDF, and the FAO — whose published work on post-harvest onion storage formed the factual foundation of this proposal, and our families and peers for their unwavering support.

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
ESP32	32-bit microcontroller with integrated Wi-Fi/Bluetooth (Espressif Systems)
GSM	Global System for Mobile Communications
SMS	Short Message Service
DHT22	Digital Humidity and Temperature sensor (model 22)
CO ₂	Carbon Dioxide
VOC	Volatile Organic Compound
RH	Relative Humidity
SQLite	Self-contained SQL database engine
API	Application Programming Interface
DFD	Data Flow Diagram
ER	Entity-Relationship (diagram)
UML	Unified Modeling Language
FPO	Farmer Producer Organisation
ICAR-DOGR	Indian Council of Agricultural Research – Directorate of Onion and Garlic Research
NHRDF	National Horticultural Research and Development Foundation
NAFED	National Agricultural Cooperative Marketing Federation of India
NCCF	National Cooperative Consumers' Federation of India
PMKSY	Pradhan Mantri Kisan Sampada Yojana
RKVY	Rashtriya Krishi Vikas Yojana
AMC	Annual Maintenance Contract
BOM	Bill of Materials
PPP	Public-Private Partnership
PLW	Physiological Loss in Weight
SaaS	Software as a Service
LTE	Long-Term Evolution (mobile network standard)

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Abstract

Onion is one of India's most widely cultivated and consumed vegetable crops, yet 30% to 40% of the harvested produce is lost every year during storage due to fungal rot, sprouting and physiological weight loss, with losses exceeding 50% during adverse weather. These losses stem primarily from high relative humidity, excess heat, poor ventilation and the time-consuming, error-prone nature of manual inspection in traditional storage structures.

This project proposes Onichan, an AI-powered IoT system that continuously monitors temperature, humidity, air quality (CO₂ and VOC) and visual condition inside an onion storage structure, and automatically maintains an optimal micro-climate by controlling exhaust fans and ventilation. An ESP32 microcontroller collects sensor and camera data and transmits it to a lightweight AI engine (built in Python) that estimates spoilage risk and predicted shelf life. Farmers and warehouse operators receive real-time information through a web/mobile dashboard and are notified of abnormal conditions through SMS alerts, even without continuous internet access.

Research by ICAR-DOGR indicates that maintaining a micro-climate of 27°C ± 2°C and 60–70% relative humidity can reduce onion storage losses from approximately 54.89% (traditional ventilated structures) to as low as 7.14%, without the very high capital and running costs of mechanical cold storage. Building on this evidence, Onichan is designed as a low-cost, solar-capable, modular system that scales from a single farmer's shed to large warehouse installations, offering a practical middle ground between low-cost traditional storage and capital-intensive refrigerated cold storage.

This report documents the problem background, literature review, system requirements, feasibility study, system design (data flow diagrams, database schema, architecture and flowchart), implementation methodology, proposed evaluation plan, and the business and policy case for adoption, including a detailed hardware cost breakdown across three deployment tiers and a comparative analysis against existing storage methods.

Status note: Onichan is at the design and bench-prototype stage. All loss-reduction and performance figures quoted in this report are design targets derived from published ICAR-DOGR research, not measurements from a completed field installation. Section 5.5 sets out the pilot validation plan required before any of these figures can be treated as measured results, and Section 7 lists the source-verification status of the literature relied upon.

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1. Introduction

1.1 Project Background

India is one of the largest producers and consumers of onion in the world, with the crop grown across nearly every state and harvested in two major seasons — Rabi (harvested March–May) and Kharif (harvested October–December). Onion is a staple ingredient in Indian households and a politically sensitive commodity: sharp price swings caused by supply shortages regularly make national headlines and affect food inflation.

A large share of these shortages is not caused by a shortfall in production but by losses that occur after harvest, while the crop is in storage. Traditional storage structures — open, naturally ventilated “chawls” or sheds — offer little control over the humidity, temperature and airflow inside the structure. High humidity encourages fungal rot, excess heat accelerates physiological weight loss, and poor ventilation traps moisture released by the bulbs themselves. Because inspection is manual, spoilage is often detected only after a significant fraction of the stored crop is already lost.

The Onichan project responds to this gap with a low-cost, AI-powered Internet of Things (IoT) system that continuously senses storage conditions, automatically intervenes through ventilation control, and keeps the farmer or warehouse operator informed in real time — combining the affordability of traditional storage with the intelligence of a monitored, semi-automated environment.

1.2 Motivation

- India loses an estimated 30% to 40% of its harvested onion crop to post-harvest spoilage every year, rising above 50% in adverse weather — a scale of loss confirmed by ICAR (Indian Council of Agricultural Research) and NHRDF studies.
- Existing low-cost storage structures give farmers no real-time visibility into the conditions inside the store; problems are usually noticed only once rot or sprouting is already visible.
- Mechanical cold storage solves the spoilage problem but is financially out of reach for most individual farmers and small cooperatives, costing upward of ₹50 lakh to build and requiring continuous high electricity consumption to run.
- Government agencies (NAFED, NCCF) themselves procure and store large volumes of onion as part of the national buffer stock, and are exposed to the same storage losses at a much larger financial scale.
- Low-cost microcontrollers (such as the ESP32), environmental sensors, and mobile connectivity have become cheap and reliable enough to make continuous, automated monitoring viable even for individual small-holder farmers.

1.3 Statement of the Problem

“Improved Onion Storage Technology for Enhancing Shelf Life of Onion” — farmers and warehouse operators currently have no affordable, automated way to monitor and control the temperature, humidity and air quality inside onion storage structures, resulting in high and often unpredictable post-harvest losses, reduced farmer income, and instability in market supply and prices.

1.4 Aims and Objectives

- To design a low-cost IoT hardware unit that continuously measures temperature, humidity, CO₂ and VOC levels, and captures periodic images of stored onion batches.
- To build a lightweight AI engine that analyses sensor and image data to estimate spoilage risk and predict remaining shelf life.
- To automatically control exhaust fans and ventilation dampers so that the storage micro-climate stays within the scientifically recommended range (approximately 25–30°C, 60–70% RH).
- To provide farmers and operators with a real-time web/mobile dashboard and SMS alerts for abnormal conditions, including in low-connectivity rural areas.
- To keep the system modular and affordable across three cost tiers, so it can scale from a single farmer's shed to a large warehouse installation.
- To evaluate the system's technical and economic feasibility, and to quantify its potential benefit to farmers, government procurement agencies and the wider agricultural supply chain.

2. Review of Literature

2.1 Related Work

Post-harvest onion storage has been studied extensively by Indian agricultural research bodies as well as international agencies, and a growing body of IoT literature has begun to apply low-cost sensing to the problem.

- ICAR-DOGR (Directorate of Onion and Garlic Research) has published field-trial data comparing storage loss across traditional ventilated structures, cold storage, and micro-climate-controlled warm storage, establishing $27^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and 60–70% RH as the recommended ambient storage band for onion.
- NHRDF's Improved Storage of Ventiladed Structures (ISVS) work documents structural design guidelines — bottom-up mesh ventilation, elevated platforms and roof insulation — that reduce passive heat and moisture build-up without any electrical intervention.
- The FAO's post-harvest guidance describes internationally used Controlled Atmosphere (CA) storage, which regulates O₂ and CO₂ levels alongside temperature and humidity, and is common in European and North American cold-chain storage.
- Recent IoT literature (e.g. ESP32-based cloud-integrated storage monitoring studies and gas-sensing / mould-detection prototypes) demonstrates that low-cost microcontrollers, humidity/temperature probes and gas sensors can be combined into a functioning remote-monitoring unit at a fraction of the cost of industrial data-logging equipment.

2.2 Comparative Analysis

The table below summarises how the three broad categories of onion storage compare on cost, loss rate and level of automation, based on ICAR-DOGR field data and typical market pricing for cold storage infrastructure.

Parameter	Traditional Ventilated Chawl	Industrial Cold Storage	Onichan (Proposed)
Initial capital cost	₹20,000 – ₹50,000	₹50 lakh – ₹1 crore+	₹9,240 – ₹35,420 per unit
Running / power cost	Negligible	Very high (continuous refrigeration)	Low (ESP32 + fans, solar-capable)
Storage loss over ~120 days	35% – 55%	~5% in storage, but rapid sprouting after removal	Target: below 10% (27°C ± 2°C, 60–70% RH)
Monitoring	Manual inspection only	Periodic manual logging / fixed thermostat	Continuous automated sensing (temp, RH, CO ₂ , VOC, images)
Automation	None	Fixed thermostat control	Automatic fan/vent control + AI-based SMS alerts
Scalability	Fixed physical layout	Rigid structural requirement	Modular — single shed to large warehouse

2.3 Limitation of the Existing System

- Traditional ventilated storage relies entirely on passive airflow and offers no way to intervene when humidity or temperature drifts outside the safe range, so losses remain high and unpredictable.
- Cold storage solves spoilage during storage but is prohibitively expensive for individual farmers, consumes large amounts of electricity, and causes onions to sprout rapidly once removed into ambient conditions, limiting its usefulness for produce headed straight to market.
- Manual inspection is labour-intensive, infrequent and subjective — by the time visible rot or sprouting is noticed, a significant fraction of the batch may already be affected.
- Existing IoT prototypes in the literature largely focus on data logging or single-parameter alerts; few combine multi-sensor environmental data, image-based spoilage detection, automatic actuation and low-connectivity SMS alerting into one affordable, farmer-usable package — the gap that Onichan is designed to fill.

3. System Analysis and Design

3.1 Requirements

3.1.1 Functional Requirements

- The system shall continuously record temperature, relative humidity, CO₂ and VOC levels from within the storage structure at a configurable interval.
- The system shall periodically capture images of stored onion batches for visual condition assessment.
- The system shall transmit sensor and image data from the microcontroller to the processing / AI layer for analysis.
- The AI engine shall estimate a spoilage-risk level (e.g. Healthy / Risk) and a predicted remaining shelf life in days.
- The system shall automatically switch exhaust fans and ventilation dampers ON or OFF based on the decision engine's output, with a manual override available to the operator.
- The system shall display real-time readings, historical trends (heat-map) and shelf-life predictions on a web/mobile dashboard.
- The system shall send an SMS alert through a GSM module when an abnormal condition (e.g. high humidity) is detected, so that operators without continuous internet access are still notified.
- The system shall log all sensor readings, images, predictions and alerts to a persistent database for later review and model improvement.

3.1.2 Non-Functional Requirements

- Affordability: total hardware cost should remain within ₹35,000 even for the high-end deployment tier, and as low as ~₹2,800–₹9,000 for budget/medium tiers (see Section 3.2.2).
- Reliability: the system should continue basic sensing and fan control even during temporary loss of internet connectivity, falling back to local (edge) logic and SMS alerts.
- Low power consumption: suitable for solar/battery-backed operation in rural areas with unreliable grid power.
- Usability: the dashboard and SMS alerts should be simple enough for a farmer with basic smartphone literacy to understand without training.
- Scalability: the same architecture should support a single small shed as well as a large multi-zone warehouse by adding more sensor nodes.
- Maintainability: components should be modular and individually replaceable (e.g. a failed sensor can be swapped without replacing the whole unit).

3.1.3 Hardware & Software Requirement

Hardware requirement (refer Section 3.2.2 for full three-tier cost breakdown):

Category	Component(s)
Compute / control	ESP32 development board (microcontroller with Wi-Fi/Bluetooth)
Temperature & humidity sensing	DHT22 (zone sensor), DS18B20 (stack probe), outside DHT22 (ambient sensor)
Air quality sensing	MQ135 (VOC sensor), MH-Z19B (CO ₂ sensor)
Vision	ESP32-CAM / OV2640 image capture module
Actuation	Relay module driving exhaust fan(s) and intake/vent dampers
Communication	Wi-Fi (dashboard) + GSM/SIM module (SMS alerts)
Power	12V power supply; battery + solar panel backup for rural/off-grid sites
Enclosure	Weatherproof (IP65) junction box for outdoor/warehouse mounting

Software requirement:

Layer	Technology
Firmware	Arduino IDE (C/C++ for ESP32)
Backend / AI engine	Python, Flask
Frontend / dashboard	HTML, CSS, JavaScript
Database	SQLite
Version control	GitHub
Alerting	SMS via GSM module (e.g. SIM800L / SIM7600 depending on tier)

3.2 Feasibility Study

3.2.1 Technical Feasibility

All hardware components specified (ESP32, DHT22/DS18B20, MQ135/MH-Z19B, relay modules, GSM modules) are widely available, well-documented, and commonly used in existing IoT and agricultural-monitoring prototypes, which substantially de-risks the hardware build. The software stack (Arduino IDE, Flask, SQLite, HTML/CSS/JS) is open-source, lightweight, and does not require specialised infrastructure, so it can run on a low-cost single-board computer or a small cloud instance. Edge-first design (with the ESP32 handling core fan control locally) ensures that the system remains functional even when internet or cloud connectivity is intermittent, which is common in rural deployment areas.

3.2.2 Economic Feasibility

The hardware is priced across three tiers so that the entry cost can be matched to the buyer's budget and scale — an individual smallholder farmer, a mid-size cooperative, or a large warehouse operator.

Component Category	Cheap / Basic Tier (₹)	Medium Tier — Recommended (₹)	High-End Tier — No-Compromise (₹)
Compute / Microcontroller	ESP32 Dev Module — 350	ESP32-WROOM-32D + I/O Shield — 650	Industrial Raspberry Pi CM4 / ESP32-S3 — 3,500
Temperature & Humidity Probes	DHT11 / DHT22 — 150	2× DHT22 + DS18B20 Stack Probe — 750	SHT31 Weatherproof + Multi-point DS18B20 Array — 2,200
Gas & Air Quality Sensors	MQ-135 — 180	MQ-135 + MH-Z19B (CO ₂) — 1,400	NDIR CO ₂ + Industrial VOC/Ammonia Probe — 4,500
Camera Module (AI vision)	OV2640 / ESP32-CAM — 450	ESP32-CAM 2MP + Night IR LED — 700	Industrial HD IP Camera, Wide-Angle — 2,800
Actuator Relays	4-Channel Relay — 180	Optocoupler 4-Ch Relay + SSR — 450	Din-Rail Industrial Relay Block — 1,200
Ventilation / Exhaust	1× 12V DC Fan — 350	2× Heavy-Duty 230V AC Exhaust Fans — 1,800	Dual Variable-Speed Industrial Blowers + Dampers — 5,500
Connectivity & Power Supply	12V Adapter + SIM800L (2G) — 650	12V 5A Supply + SIM7600 (4G) — 1,850	Industrial PSU + Dual-SIM 4G Gateway — 4,200
Power Backup	Not included — 0	12V 7Ah Lead-Acid Battery — 1,200	LiFePO ₄ Battery + Solar Charge Controller — 6,500
Enclosure & Wiring	Plastic Box & Jumper Wires — 250	IP65 Weatherproof Enclosure — 600	Flame-Retardant Metallic Enclosure — 1,800
Base Hardware Total	2,560	8,400	32,200
Buffer / Unaccounted Parts (10%)	256	840	3,220
Total Bill of Materials (BOM)	2,816	9,240	35,420

Annual maintenance and calibration cost is estimated at approximately ₹500/year (Basic), ₹1,200/year (Medium, covering sensor calibration and software patches) and ₹3,500/year (High-End, including preventive maintenance and priority support).

Revenue Model

- Direct hardware sales: Medium units to individual farmers/FPOs at an indicative ₹14,999 per kit; High-End units to commercial warehouses at an indicative ₹55,000 per unit.
- Software-as-a-Service (SaaS): a subscription of ₹199–₹499 per month per unit for cloud analytics, historical heat-maps, predictive shelf-life modelling and priority SMS alerting.
- Enterprise / B2G annual maintenance contracts (AMC), typically 10–15% of the total project cost, for large-scale warehouse or buffer-stock deployments.

These figures are illustrative planning estimates derived from current retail component pricing and comparable SaaS/AMC benchmarks; they should be validated against final Bill-of-Materials quotations and market pricing before commercial launch. Open questions that must be tested before this becomes a firm business plan include: actual component cost at procurement volume; assembly, testing and installation cost per unit; the channel cost of reaching dispersed rural customers; farmers' ability to finance the upfront hardware price; competition from low-technology alternatives (shading, improved passive ventilation, simple thermometers); and whether annual billing aligned to the rabi storage cycle fits customers better than monthly SaaS billing.

3.3 Project Design

3.3.1 DFD / UML Diagrams

The Level-0 (context) diagram below shows the system as a single process exchanging data with its external entities — the sensor/camera array, the farmer or warehouse operator, the actuators and the GSM/SMS gateway.

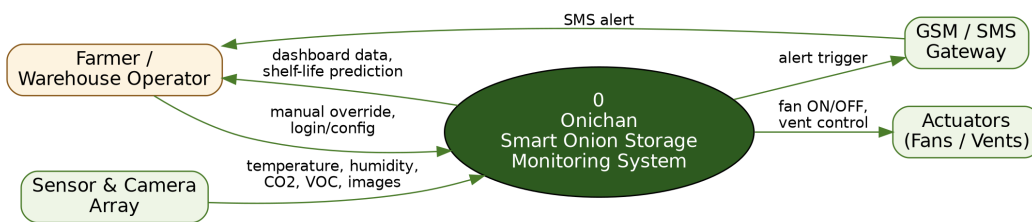


Fig. 3.1 — Level-0 Data Flow Diagram (Context Diagram)

The Level-1 diagram decomposes the system into its five core processes — data acquisition, AI analysis, decision-making, actuator control, and the alert/dashboard service — along with the two data stores they read from and write to.

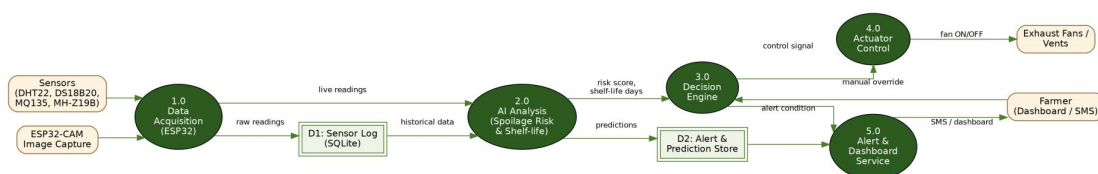


Fig. 3.2 — Level-1 Data Flow Diagram

3.3.2 Database Schema

Sensor readings, captured images, AI predictions and alerts are each logged against the storage unit that produced them, and every storage unit is linked to the user (owner/operator) responsible for it. This normalised schema keeps historical data queryable for both real-time dashboard rendering and later AI-model retraining.

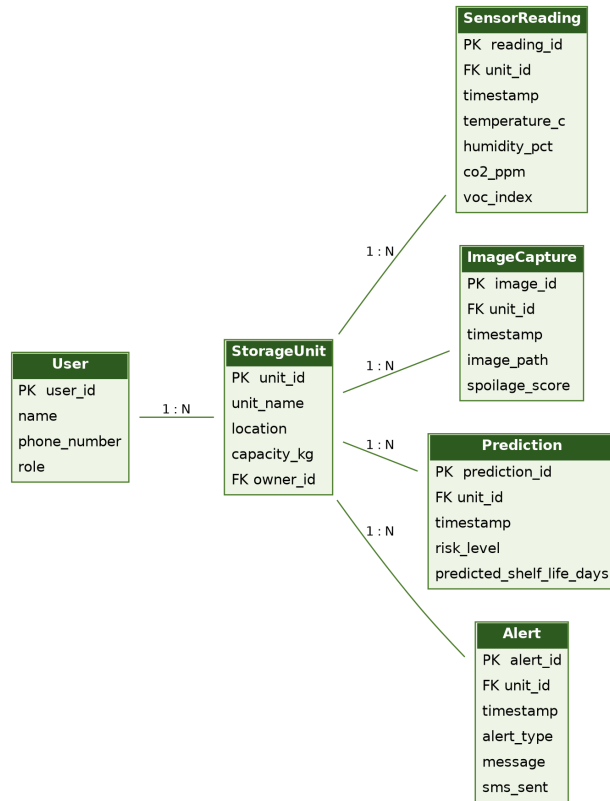


Fig. 3.3 — Database (ER) Schema (SQLite)

3.3.3 Flow Chart

The operational cycle of the system follows a repeating ten-step loop, from raw data collection through to the farmer receiving an actionable decision.

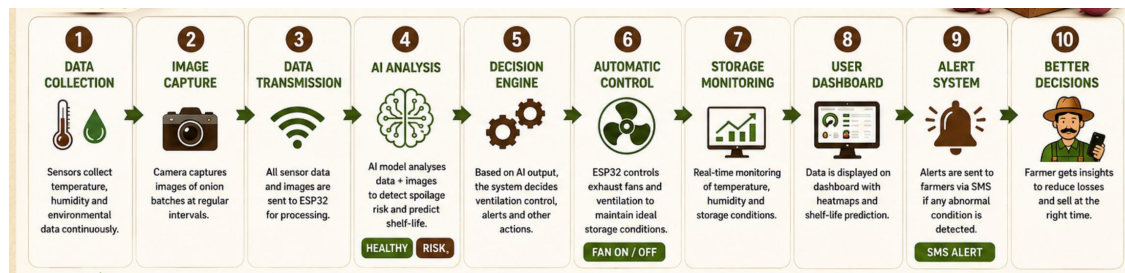


Fig. 3.4 — Process Flow Chart (10-step operational cycle)

3.3.4 Architectural Diagram

At the architectural level, the ESP32 microcontroller bridges the sensing/vision hardware and the AI engine. The AI engine (in Python) reads and writes to the SQLite database, issues fan-control decisions back to the ESP32, pushes real-time data to the dashboard, and triggers SMS alerts through the GSM module.

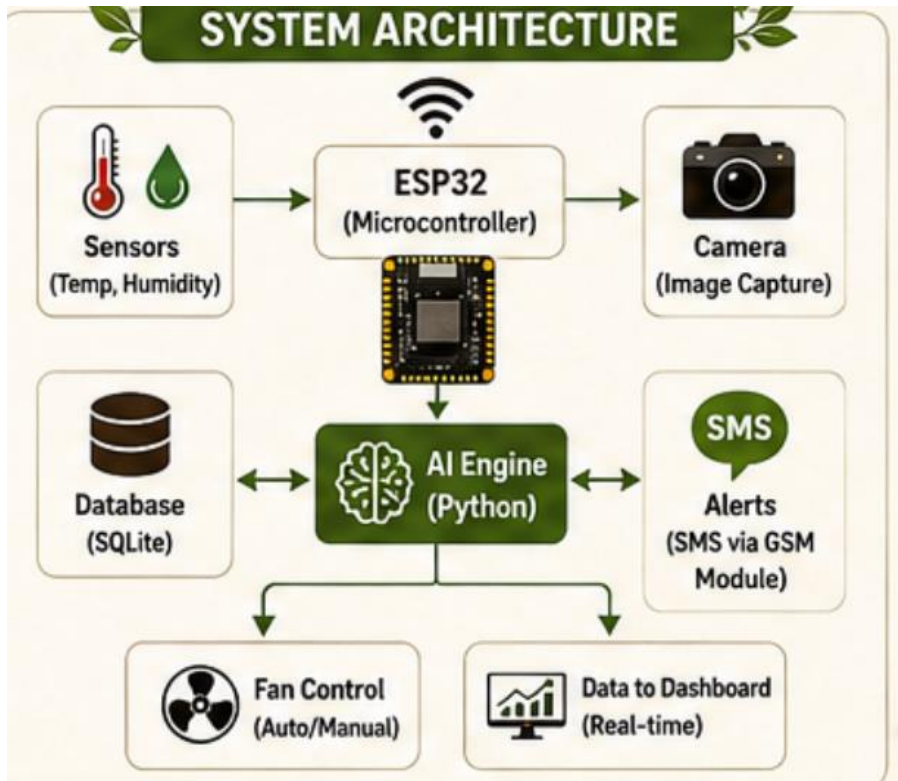


Fig. 3.5 — System Architectural Diagram

3.4 Gantt Chart

The indicative project timeline below sequences requirement gathering, design, hardware procurement, firmware/AI/backend development, integration, testing, field trial and final documentation across roughly fifteen weeks. Actual dates will be adjusted to the hackathon and institutional deadlines.

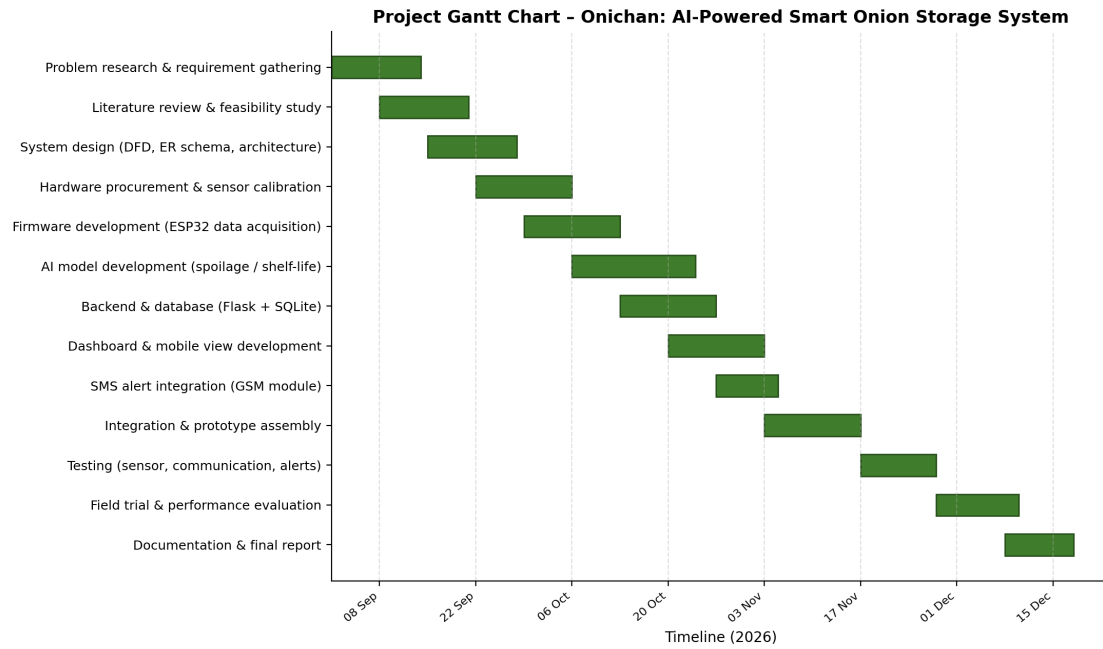


Fig. 3.6 — Project Gantt Chart

4. Methodology & Implementation

4.1 Block Diagram of Project

The project is organised into three functional layers: a sensing layer (sensors and camera), a processing & storage layer (the ESP32 paired with a Python/Flask AI engine and an SQLite database), and an actuation & output layer (fans/dampers, dashboard, and SMS alerts).

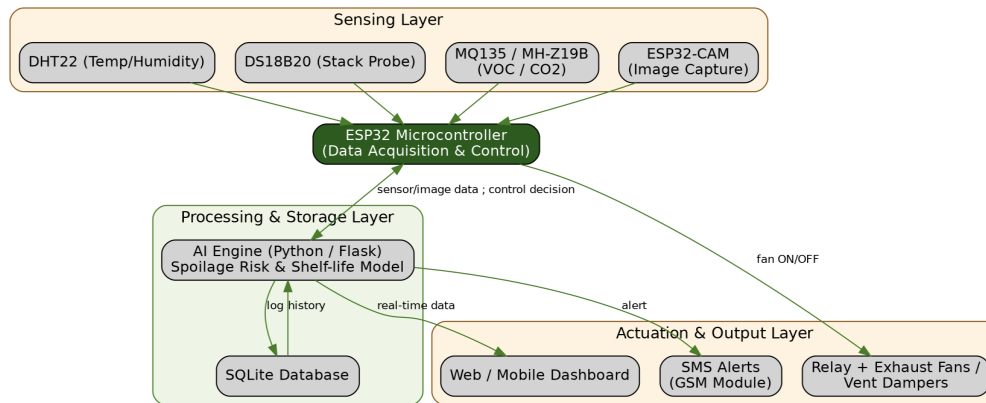


Fig. 4.1 — Block Diagram of the Project

4.2 Tools & Technologies Used

Tool / Technology	Purpose
Arduino IDE	Firmware development and flashing for the ESP32 microcontroller
Python	AI engine — spoilage-risk classification and shelf-life prediction logic
Flask	Lightweight backend web server exposing sensor data and predictions via API
HTML / CSS / JavaScript	Web dashboard front-end (desktop and mobile-responsive views)
SQLite	Embedded relational database for sensor logs, images metadata, predictions and alerts
GitHub	Source-code version control and team collaboration
GSM Module (SIM800L / SIM7600)	SMS alert delivery independent of Wi-Fi/internet availability

4.3 Development Environment

- **Firmware:** Arduino IDE targeting the ESP32 board package, programmed in C/C++.
- **Backend:** Python 3 with the Flask micro-framework, running on a low-cost single-board computer or small cloud VM.
- **Database:** SQLite file-based database, chosen for its zero-configuration setup and suitability for edge/embedded deployment.
- **Frontend:** Static HTML/CSS/JavaScript dashboard served by Flask, designed mobile-first for use on a farmer's smartphone.

- Version control & collaboration: Git and GitHub, with the codebase organised into firmware, backend and frontend modules.

4.4 Key Features Implemented

- Continuous monitoring of temperature, humidity and air quality inside the storage structure.
- AI-based spoilage-risk prediction that analyses sensor and image trends to flag “Healthy” vs. “Risk” conditions and estimate remaining shelf life.
- Automatic environmental control — exhaust fans and fresh-air intake are switched on or off to maintain the recommended humidity and temperature band.
- Instant SMS and dashboard alerts whenever an anomaly or risk condition is detected.
- Remote monitoring and manual override through a web/mobile dashboard, so an operator can intervene directly if required.

The table below states the current implementation status of each feature honestly, since the analytics-heavy features depend on field data that can only be collected during pilots (Section 5.5).

Feature	Description	Status
Multi-point temperature & RH sensing	DHT22/DS18B20-class or SHT31 probes distributed through the onion stack	Prototype
CO ₂ / VOC monitoring	MH-Z19B NDIR CO ₂ sensor and MQ-135-class VOC sensor as early indicators of rot/mould	Prototype
Automated ventilation control	Relay-driven exhaust fans and intake vents on threshold/timing logic	Prototype
Edge autonomy	Control logic runs locally on the ESP32; outages do not interrupt ventilation	In design
Cloud dashboard & heat-maps	Storage-wise temperature/humidity trends with spatial heat-maps	In design
AI-based spoilage alerts	Trend-based early warning of sprouting/rot via app and SMS	Concept — needs pilot data
Predictive spoilage engine	Longer-horizon risk scoring of storage lots	Concept — needs pilot data
Camera-based visual check	Optional ESP32-CAM / IP camera for remote inspection	Optional add-on
Power backup / solar option	Lead-acid backup (field tier); LiFePO ₄ + solar controller (industrial tier)	In design

4.5 System Interface

The proposed dashboard shows live temperature, humidity, storage status and a colour-coded heat-map alongside the AI's predicted shelf life. The same information is mirrored in a mobile-friendly view, and abnormal conditions additionally trigger an SMS alert so the farmer is notified even without opening the app.

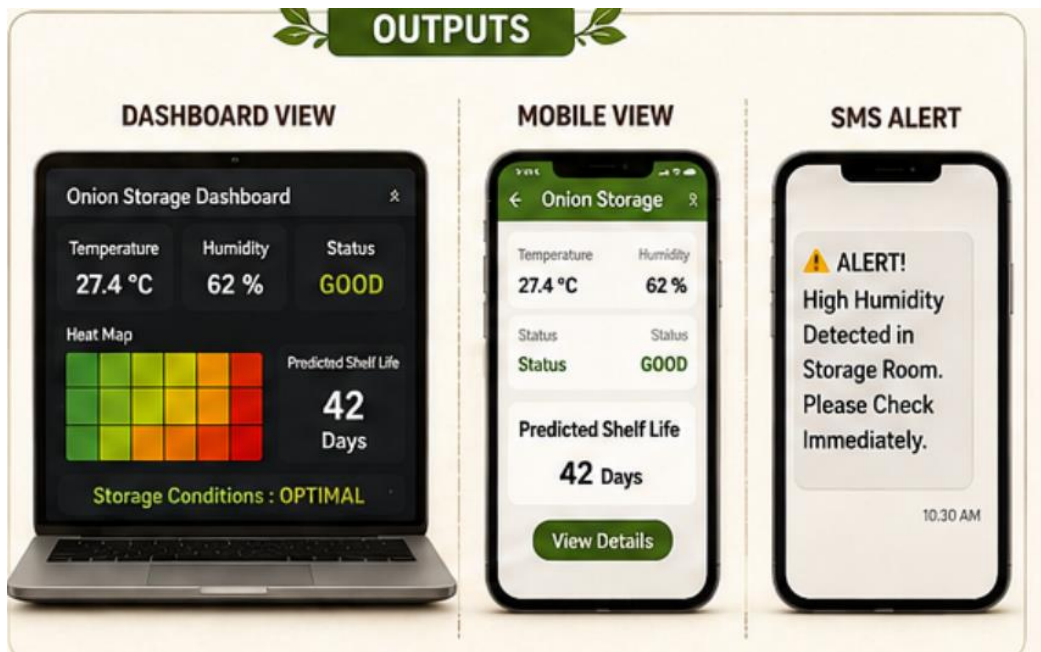


Fig. 5.1 — Proposed dashboard, mobile view and SMS alert interfaces

The hardware overview below summarises the controller, sensors, and actuator/indicator components that make up a single deployed unit.



Fig. 4.2 — Hardware Overview

5. Results & Analysis

Onichan is being submitted at the idea / prototype-planning stage of Smart India Hackathon 2026. The material in this chapter therefore presents the proposed interface design, the performance targets the team intends to validate, and the test plan that will be executed once the hardware prototype is assembled — rather than results from a completed field deployment. This distinction is stated explicitly so the report accurately reflects the current stage of development.

5.1 Proposed Interface Design (GUI)

5.1.1 Dashboard

The web dashboard (see Fig. 5.1) is designed to show live temperature and humidity, an overall status indicator (e.g. GOOD / RISK), a colour-coded heat-map of recent readings, and the AI-predicted shelf life in days, alongside an overall “storage conditions” summary.

5.1.2 Mobile View

A responsive mobile view mirrors the dashboard's key metrics — temperature, humidity, status and predicted shelf life — in a compact layout intended for quick daily checks by a farmer on a basic smartphone.

5.1.3 Alerts & Reports

When a sensor reading crosses a configured threshold (e.g. relative humidity above 75%), the system is designed to immediately push an SMS alert (“High Humidity Detected in Storage Room. Please Check Immediately.”) in addition to updating the dashboard, so that operators without an active data connection are still informed.

5.2 Target Performance Metrics

The following targets are derived from ICAR-DOGR's published benchmark figures for micro-climate-controlled onion storage and will be used as the success criteria during field trials of the completed prototype.

Metric	Baseline (Traditional Storage)	Onichan Target
Storage loss over ~120 days	35% – 55%	Below 10%
Post-storage sprouting after removal	Moderate to High	Minimal (ambient-acclimated storage)
Sensor-to-alert latency	Not applicable (manual only)	Under 60 seconds from threshold breach to SMS/dashboard alert
Storage micro-climate maintained	Uncontrolled	27°C ± 2°C, 60–70% RH for at least 90% of storage duration
Manual inspection effort	Daily manual checks	Reduced to periodic spot-checks / exception-based only

5.3 Test Plan

The following test cases are planned for validation once the hardware prototype and firmware are assembled. Results (pass/fail, measured values) will be recorded against this plan during the testing phase shown in the Gantt chart (Section 3.4).

5.3.1 Sensor and Detection Test Case

Test ID	Description	Expected Result
T-01	Verify DHT22 temperature/humidity readings against a calibrated reference thermometer/hygrometer	Reading within $\pm 0.5^{\circ}\text{C}$ / $\pm 2\%$ RH of reference instrument
T-02	Verify MQ135/MH-Z19B gas sensor response to a controlled CO_2 /VOC source	Sensor output changes proportionally and stabilises within the expected settling time
T-03	Verify ESP32-CAM image capture at scheduled intervals under varying storage lighting	Images captured, timestamped and stored without gaps

5.3.2 Communication and Dashboard Test Case

Test ID	Description	Expected Result
T-04	Verify ESP32-to-backend data transmission over Wi-Fi under normal conditions	Sensor data appears on the dashboard within 5 seconds of capture
T-05	Verify system behaviour during a simulated Wi-Fi outage	ESP32 continues local fan-control logic; data buffers and syncs once connectivity resumes
T-06	Verify dashboard heat-map and shelf-life prediction update correctly as new readings arrive	Displayed values match the latest logged database entries

5.3.3 Alert and System Performance Test Case

Test ID	Description	Expected Result
T-07	Trigger a simulated high-humidity condition and measure time to SMS alert	SMS received within 60 seconds of threshold breach
T-08	Verify automatic fan activation when temperature/humidity exceeds the configured band	Relay switches fan ON within the decision-engine's polling interval, and OFF once conditions normalise
T-09	Run the system continuously for 72 hours to check for memory leaks, sensor drift or communication drop-outs	No crashes, reboots or unexplained data gaps over the test period

5.4 Comparative Analysis with Existing Systems

Revisiting the comparison introduced in Section 2.2, Onichan is positioned as a middle ground: it costs an order of magnitude less than industrial cold storage while offering continuous automated monitoring and control that traditional ventilated chawls cannot provide.

Parameter	Traditional Chawl	Industrial Cold Storage	Onichan (Proposed)
Setup cost	₹20,000–₹50,000	₹50 lakh+	₹9,240–₹35,420
Running cost	Negligible	Very high	Low (solar-capable)
Loss rate	35%–55%	~5% (rapid sprouting after removal)	Target <10%
Automation & alerts	None	Manual monitoring	Automated fan control + AI-based SMS alerts

5.5 Key Risks and Validation Plan

The team is deliberately transparent about the principal risks to the design and the commercial model, and the mitigations planned for each.

Risk	Why it matters	Proposed mitigation
Sensor drift/failure in humid, dusty stores	False alarms, or missed spoilage conditions	Industrial-grade sensors on higher tiers; scheduled calibration; validation during pilots
Poor rural connectivity	Cloud alerts and dashboards fail	Edge autonomy on-device; SMS fallback; battery/solar backup
Farmer financing constraints	Upfront cost blocks adoption	Subsidy alignment (PMKSY, RKVY-RAFTAAR); FPO aggregation; subscription pricing
Overstated loss-reduction claims	Loss of trust if field results miss targets	Conservative targets; independent pilot measurement before performance claims
Service reach beyond home region	AMC promises require technicians on call	Regional rollout starting in Maharashtra's onion belt; local technician training
Substitute low-tech practices	Improved passive ventilation may deliver part of the benefit more cheaply	Position the product on continuous monitoring, automation and data, not hardware alone

Before any of the loss-reduction figures in this report can be claimed as measured results, the team proposes the following validation sequence:

- Bench validation of sensor accuracy and response time against reference instruments, including MQ-135 cross-sensitivity checks in store-like air.
- Retrofit of one traditional chawl and one FPO godown with field-tier units, logging one full rabi storage season.

- Measurement of rotting, sprouting and physiological loss in weight (PLW) against a matched control structure without automation, to quantify the actual loss reduction achieved.
- Refinement of threshold and alert logic with ICAR-DOGR guidance, and end-to-end validation of the alert pipeline.
- Collection of operational telemetry (energy use, battery/solar performance, component failures) to convert O&M and AMC estimates into priced offers.
- Usability trials of the app and SMS interface with farmers and FPO staff in local languages.

The system cannot compensate for poor curing, damaged bulbs or badly built storage sheds — storage outcomes remain jointly determined by agronomy and engineering. The pilots are designed to quantify what share of current losses the system can realistically remove.

6. Conclusion and Future Scope

6.1 Conclusion

Post-harvest onion storage losses of 30–40% (and over 50% in adverse conditions) represent one of the most tractable inefficiencies in India's agricultural supply chain, because the root cause — uncontrolled temperature and humidity in storage — is well understood and, per ICAR-DOGR research, can be substantially mitigated by holding storage conditions at $27^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and 60–70% relative humidity. Onichan translates this research finding into a low-cost, modular IoT product: an ESP32-based sensing and actuation unit, a lightweight AI engine for spoilage-risk and shelf-life prediction, and a dashboard/SMS alerting layer that keeps farmers informed even in low-connectivity areas.

Priced from roughly ₹2,800 for a basic prototype tier up to ₹35,000 for an industrial-grade unit, the system is designed to be accessible to individual farmers, cooperatives, and government procurement agencies alike, while remaining dramatically cheaper to build and run than mechanical cold storage. If validated at field scale, the approach has the potential to meaningfully reduce national post-harvest losses, stabilise onion market prices, and improve farmer incomes without requiring large capital investment.

6.2 Future Scope

6.2.1 Limitations

- The system currently targets ambient micro-climate control and cannot achieve the near-zero spoilage of true refrigerated cold storage, which remains preferable for very long-term or export-grade storage needs.
- Performance figures in this report are drawn from published ICAR-DOGR research and component-level specifications; they have not yet been validated on a completed Onichan field prototype, and real-world results may vary by climate zone, onion variety and storage structure.
- GSM-based SMS alerting depends on mobile network coverage, which can be patchy in some remote storage locations.
- The AI spoilage/shelf-life model will initially rely on a limited training dataset and will need to be retrained and validated across multiple onion varieties (Rabi vs. Kharif) and climatic regions before being trusted for large-scale decision-making.

6.2.2 Potential Enhancements

- Integration with weather forecasting APIs to pre-emptively adjust ventilation ahead of predicted humidity spikes.
- Continuous improvement of the AI model using accumulated historical storage data across many deployed units.
- Expansion of the sensing and control logic to other post-harvest crops such as potato and garlic, which face similar storage challenges.
- Integration with agricultural supply-chain and marketplace platforms so that predicted shelf life can inform selling decisions directly.

- Enabling onion-storage telemetry as verifiable collateral data for micro-loans through rural banking / FinTech partnerships.

6.2.3 Scalability Aspects

- The modular, three-tier hardware design allows the same core architecture to scale from a single household pantry unit (~₹2,500 BOM) to a multi-zone warehouse installation with multiple sensor nodes feeding one central AI engine.
- A Public-Private Partnership (PPP) model — in which government schemes such as PMKSY or RKVY subsidise 50–75% of hardware cost while the company charges a recurring SaaS/AMC fee — could accelerate adoption among smallholder farmers who cannot pay the full upfront cost.
- The same off-grid, solar-capable design is well suited to export in tropical, grid-unstable regions of South Asia, Africa and the Middle East that face similar post-harvest storage challenges.

7. References

The sources below were consulted while preparing this report. Citations were recorded from the team's working notes and are incomplete; each must be verified against the original publication before final academic submission. No source has been invented for this document.

Source (as recorded in working notes)	Used for	Status
ICAR-DOGR research paper on storage-loss evaluation and comparison of storage structures (title, authors, year, journal not recorded)	120-day loss figures (Sections 2.2, 5)	To be verified
Agrovent article, "Should India change its onion storage culture?" (web resource; exact URL/date not recorded)	Microclimate storage analysis; $27\pm 2^{\circ}\text{C}$ / 65% RH design basis	To be verified
IoT-based controlled-environment storage with gas and mould detection (arXiv preprint; authors/identifier not recorded)	Gas-sensing and control approach	To be verified
IoT-based cloud-integrated smart onion storage (IJCS PUB; volume/authors not recorded)	ESP32 cloud-integration precedent	To be verified
Component price estimates	Cost tables (Section 3.2.2)	Desk estimates from online retailers; re-quote at procurement
Buffer-stock procurement figures (NAFED / NCCF, >2.00 LMT)	Government benefits (Section 1.2)	To be re-verified against current Ministry of Consumer Affairs data

Industry References

- National Onion Association (USA) — industry data on onion storage practices.
- Ministry of Agriculture & Farmers Welfare, Government of India — policy and procurement context.
- Agricoop (Department of Agriculture & Farmers Welfare) — agricultural statistics and schemes.

Standards & Technical References

- Espressif Systems — ESP32 Series Datasheet and Technical Reference Manual (official documentation).
- Arduino — Official Arduino IDE and language reference documentation.
- Flask — Official Flask (Python) web framework documentation.
- Scikit-learn — Official documentation for machine-learning model development.

Research Papers

- "Post-Harvest Storage of Onion", PubMed Central (PMC) — review of onion storage physiology and loss mechanisms.
- IEEE Xplore — Smart Agriculture & IoT literature on sensor-based environmental monitoring.

- MDPI — Agricultural storage systems research on controlled-environment storage.
- IJCSPUB — “IoT-Based Cloud-Integrated Improved Onion Storage”, detailing ESP32-based multi-sensor integration.
- arXiv / ResearchGate — “IoT-Based Controlled Environment Storage with Gas Sensing and Mould Detection”, exploring low-cost prototype hardware for post-harvest loss reduction.

Agricultural Organisations & Government Sources

- ICAR (Indian Council of Agricultural Research) — national storage-loss research, including ICAR-DOGR field trials on micro-climate-controlled warm storage ($27^{\circ}\text{C} \pm 2^{\circ}\text{C}$, 60–70% RH reducing loss from ~54.89% to ~7.14% over 120 days).
- NHRDF (National Horticultural Research and Development Foundation) — Improved Storage of Ventilated Structures (ISVS) design guidelines.
- FAO (Food and Agriculture Organization of the United Nations) — international post-harvest loss and Controlled Atmosphere (CA) storage guidance.
- NAFED (National Agricultural Cooperative Marketing Federation of India) and NCCF (National Cooperative Consumers' Federation of India) — national onion buffer-stock procurement and storage context.

Note: this reference list reflects the sources and organisations consulted during the desk research for this proposal, drawn from the team's Smart India Hackathon idea submission and supporting research notes. Full citations (author, publication year, exact URL/DOI) should be added for each source before final academic submission, per the institution's referencing style.